Ordinal-sum factorization for polar preorder polytopes

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Abstract

We prove that the h^*-polynomial of the polar reflexive preorder polytope factors under ordinal sums, as conjectured by Athanasiadis and Chapoton. An explicit piecewise linear bijection of integer lattices makes the corresponding Minkowski functionals additive. Their lattice generating functions, and hence their h^*-polynomials, therefore multiply.

Task 1. Full solution

For a preorder τ on a nonempty finite set E, let

$$K_\tau = \operatorname{conv}\big(\{-e_i : i \in E\} \cup \{\mathbf{1}_I : \varnothing \neq I \text{ an order ideal of } \tau\}\big). \tag{1}$$

Here e_i is a standard basis vector and $\mathbf{1}_I$ is the indicator vector of I. By [1, Section 6], this is precisely the polar polytope $\mathcal{R}_\tau^\vee$ in the question. An ordinal sum $\tau_1 \oplus \tau_2$ on disjoint sets E_1, E_2 retains each internal preorder and declares every element of E_1 smaller than every element of E_2.

Theorem 1. *For arbitrary finite preorders τ_1, τ_2,*

$$\boxed{h^*(K_{\tau_1 \oplus \tau_2}, t) = h^*(K_{\tau_1}, t) h^*(K_{\tau_2}, t).}$$

Thus Conjecture 6.3 of [1] holds.

An explicit Minkowski functional

The order polytope in the decreasing convention is

$$O_\tau = \{u \in [0,1]^E : u_i \geq u_j \text{ whenever } i \leq_\tau j\}.$$

It equals the convex hull of the indicators of all order ideals, including the empty ideal. For completeness, for any $u \in O_\tau$ each nonempty superlevel set $\{i : u_i \geq a\}$ is an ideal. Writing u as the sum of its successive level differences times these indicators, and adding the remaining coefficient at 0, gives the required convex combination. The reverse inclusion is immediate.

The origin is in the interior of K_τ: it is a convex combination with strictly positive coefficients of the affinely independent vertices $-e_i$ and $\mathbf{1}_E$. Consequently K_τ is full dimensional, and

$$K_\tau = \operatorname{conv}(O_\tau \cup (-\Delta_E)), \qquad \Delta_E = \operatorname{conv}(0, e_i : i \in E).$$

For $z \in \mathbb{R}^E$, define

$$M(z) = \max(\{0\} \cup \{z_i : i \in E\}), \qquad U_i(z) = \max(\{0\} \cup \{z_j : i \leq_\tau j\}).$$

Let $g_\tau(z) = \inf\{k \geq 0 : z \in kK_\tau\}$ be its Minkowski functional, also called its gauge.

Lemma 1. *For every $z \in \mathbb{R}^E$,*

$$g_\tau(z) = M(z) + \sum_{i \in E} U_i(z) - \sum_{i \in E} z_i. \tag{2}$$

In particular, g_τ is integer valued on $\mathbb{Z}^E$.

Proof. The convex-hull description says that $z \in kK_\tau$ exactly when $z = u - v$, where u is nonnegative and weakly decreasing along τ, v is nonnegative, and

$$\max_i u_i + \sum_i v_i \le k.$$

Indeed, u belongs to $(\max_i u_i)O_\tau$ and $-v$ belongs to $(\sum_i v_i)(-\Delta_E)$; this proves one direction. Conversely a convex combination from the two polytopes gives such a decomposition with these two quantities bounded by its respective coefficients times k.

Equivalently, minimize $\max_i u_i + \sum_i (u_i - z_i)$ among nonnegative decreasing vectors u with $u \ge z$ coordinatewise. The coordinatewise least such vector is $U(z)$. Reflexivity gives $U_i(z) \ge z_i$, and transitivity gives $U_i(z) \ge U_j(z)$ when $i \le_\tau j$. Every other feasible u satisfies $u_i \ge u_j \ge z_j$ for all j above i, so $u_i \ge U_i(z)$. The objective is nondecreasing in every coordinate of u and therefore takes its minimum at $U(z)$. Finally $\max_i U_i(z) = M(z)$, proving (2). □

The lattice bijection for an ordinal sum

Put $\tau = \tau_1 \oplus \tau_2$ and write $z = (x, y)$ according to $E_1 \sqcup E_2$. Set $c = M(y)$. Define

$$\Phi(x, y) = (x - c\mathbf{1}_{E_1}, y). \tag{3}$$

This is a bijection of $\mathbb{Z}^{E_1} \times \mathbb{Z}^{E_2}$, with inverse $(x', y) \mapsto (x' + M(y)\mathbf{1}_{E_1}, y)$. We claim the stronger identity

$$g_\tau(x, y) = g_{\tau_1}(x - c\mathbf{1}_{E_1}) + g_{\tau_2}(y) \qquad (x, y \text{ real}). \tag{4}$$

Write $U_i^1(x)$ and $U_j^2(y)$ for the maxima within the two preorders. The maxima for the ordinal sum are

$$U_i(x, y) = \max(U_i^1(x), c) \quad (i \in E_1), \qquad U_j(x, y) = U_j^2(y) \quad (j \in E_2).$$

Since $c \ge 0$, with $a_+ = \max(a, 0)$ one also has

$$U_i^1(x - c\mathbf{1}) = (U_i^1(x) - c)_+, \qquad M(x - c\mathbf{1}) = (M(x) - c)_+.$$

If $p = |E_1|$, formula (2) now gives

$$\begin{aligned} g_{\tau_1}(x - c\mathbf{1}) &= (M(x) - c)_+ + \sum_{i \in E_1} (U_i^1(x) - c)_+ - \sum_{i \in E_1} x_i + pc \\ &= \max(M(x), c) - c + \sum_{i \in E_1} \max(U_i^1(x), c) - \sum_{i \in E_1} x_i. \end{aligned}$$

Adding $g_{\tau_2}(y) = c + \sum_j U_j^2(y) - \sum_j y_j$ is exactly (2) for τ, proving (4).

Multiplication of the generating functions

Define the formal series

$$\Theta_\tau(t) = \sum_{z \in \mathbb{Z}^E} t^{g_\tau(z)}.$$

Its exponents are nonnegative integers. Its coefficients are finite because $\{z : g_\tau(z) \le k\} = kK_\tau$ is bounded. The bijection (3) and identity (4) imply

$$\Theta_{\tau_1 \oplus \tau_2}(t) = \Theta_{\tau_1}(t)\Theta_{\tau_2}(t).$$

For $n = |E|$, summing over all dilates containing each lattice point gives

$$\sum_{k \ge 0} \#(kK_\tau \cap \mathbb{Z}^E)t^k = \frac{\Theta_\tau(t)}{1-t} = \frac{h^*(K_\tau, t)}{(1-t)^{n+1}}.$$

Thus $\Theta_\tau(t) = h^*(K_\tau, t)/(1-t)^n$. Substitution into the multiplicative identity proves Theorem 1.□

The proof allows equivalent distinct elements in either preorder. It also extends to an empty factor by assigning it the one-point polytope and polynomial 1. The map Φ is linear with an integer matrix of determinant 1 on each region where $M(y)$ is a fixed coordinate or zero. The argument requires this piecewise linear lattice bijection, not a global linear equivalence.

Verification and attribution

The polar vertex description and the conjecture are due to Athanasiadis and Chapoton [1]. The explicit gauge formula and lattice bijection supply the proof. The verifier compares the gauge with independently enumerated rational facets, then checks the bijection and full h^*-factorization for every ordinal pair of total size at most four. Larger examples include equivalent elements. All four exact verification groups pass.

All 26 pages of the current primary were read. Bounded checks found no earlier full target claim or identified journal final. The related support-enumerator duality and reflexive-polytope word formulas concern different statements, as recorded in the literature audit. Review is by the authoring agent only, without independent peer review or proof-assistant certification.

Tasks 2–5. Proposed follow-up directions

Task 2. Use the piecewise linear bijection to transport lattice triangulations between the ordinal-sum polytope and the free sum of the two polar polytopes.

Task 3. Extend the gauge formula to weighted ideal indicators and determine which integral weights preserve ordinal-sum factorization.

Task 4. Find analogous lattice bijections for other operations on preorders, including disjoint unions and substitutions at a vertex.

Task 5. Study when the factorization of $h^*(K_\tau, t)$ detects an ordinal-sum decomposition of the underlying preorder.

References

[1] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1, 2026.